ECG Interpretation

Dr. Muhammad Reihan, Internal Medicine | Source: `ECG.pdf` (121 slides)

> **Legend:** ⭐ = high-yield · *[added]* = not on slides · 🔎 **my reading** = my interpretation of an **unlabelled** practice strip. The lecturer explained these live; check them against your own class notes.
>
> **How this deck is built:** about 90 of the 121 slides are ECG images with only a heading ("Voltage", "Axis", "Rhythm", "P-wave abnormalities"…). The lecturer walks through the **same 11-step comment** for every strip. These notes keep the strips in order, and where a strip has no caption I give my reading.
>
> 🔗 **Bridge from earlier lectures (same lecturer):**
>
> - **Cardiac arrhythmias:** the rhythm strips (AF, flutter, SVT, AV blocks, VT, VF) were covered there; this lecture teaches how to **read** them.
> - **CAD:** ST elevation by territory and reciprocal change.
> - **Pericardial:** diffuse ST↑ + PR↓, low voltage.
> - **HTN / valvular:** LVH voltage criteria, P mitrale (MS), P pulmonale (pulmonary HTN).

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1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	⭐ Describe **in sequence** the steps of ECG interpretation	The 11-step list
Knowledge	Identify rhythm, site of origin, causes and significance of ECG changes	Strip → diagnosis
Skill	⭐ Apply a **systematic approach** to a 12-lead ECG	Comment on an ECG
Skill	⭐ Distinguish **normal** from common **abnormal** findings	Normal values

Main subtopics: basics (conduction system, ECG principles) · leads (limb/chest, bipolar/unipolar) · systematic interpretation: identification → standardisation → voltage → axis → heart rate → P wave → PR → QRS → ST → T.

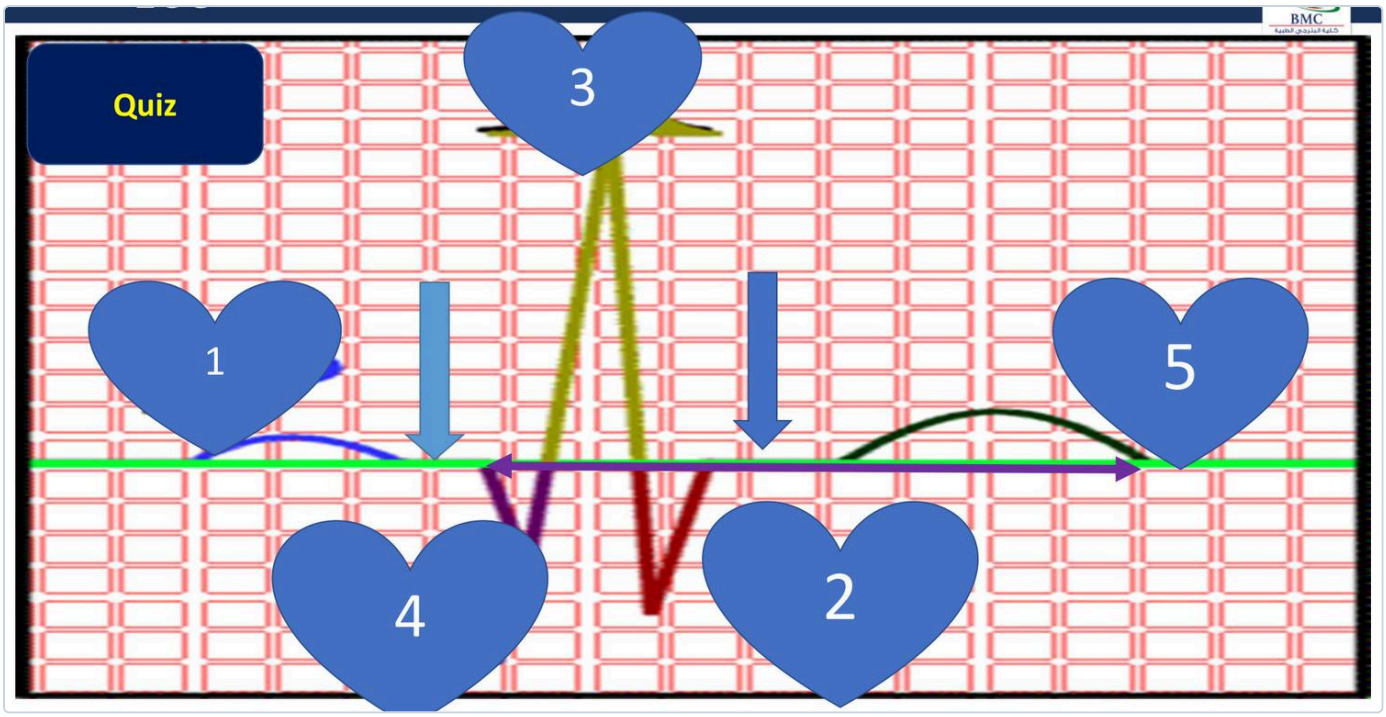
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2. Main Content (Lecture Order)

2.1 Opening Quiz & History (Slides 4–6)

Slide 4 quiz: label hearts 1–5 on a PQRST complex. *[added]* Read left to right, they are 1 = **P wave**, 4 = **Q**, 3 = **R**, 2 = **S**, 5 = **T wave**.

- **Einthoven 1893** (the first "EKG", 1901) → Nobel Prize 1924 *[added]*.



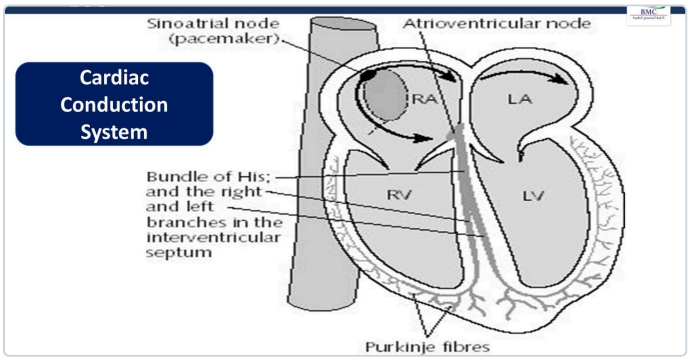
Slide 4: PQRST quiz

2.2 Cardiac Conduction System (Slides 7–8) ⭐

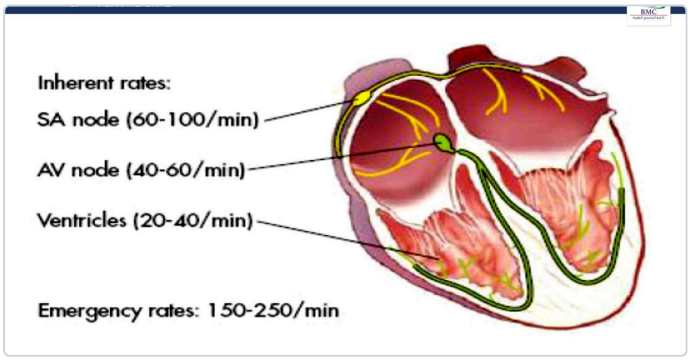
SA node (pacemaker) → atria → **AV node** → **bundle of His** → **right and left bundle branches** in the septum → **Purkinje fibres**.

⭐ The "fail-safe" system: inherent (escape) rates

Pacemaker	Rate
SA node	**60–100/min**
AV node (junctional)	**40–60/min**
Ventricles	**20–40/min**
Emergency rates (ectopic tachycardias)	**150–250/min**



Slide 7: conduction system



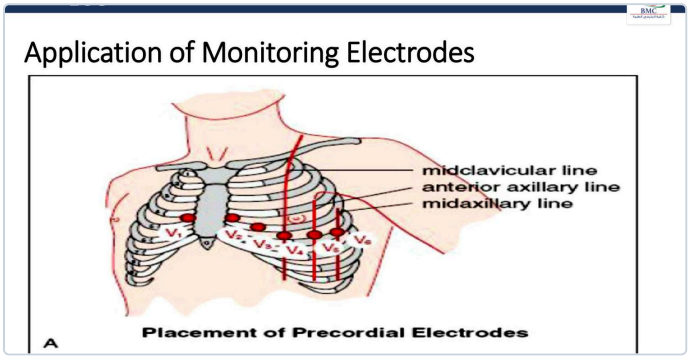
Slide 8: inherent rates ("fail-safe")

> **Use:** in complete heart block, an escape at **40–60** with a narrow QRS = **junctional**; **20–40** with a wide QRS = **ventricular** (worse).

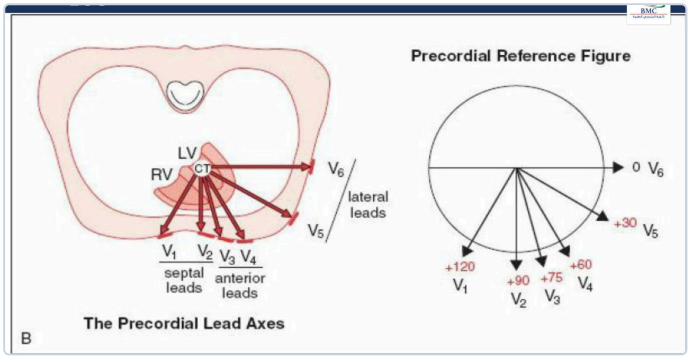
2.3 Leads (Slides 9–16) ⭐

Group	Leads	Type
Limb leads	**I, II, III**	**Bipolar** (Einthoven's triangle)
	aVR, aVL, aVF	**Unipolar** (augmented)
Chest (precordial) leads	**V1–V6**	**Unipolar**

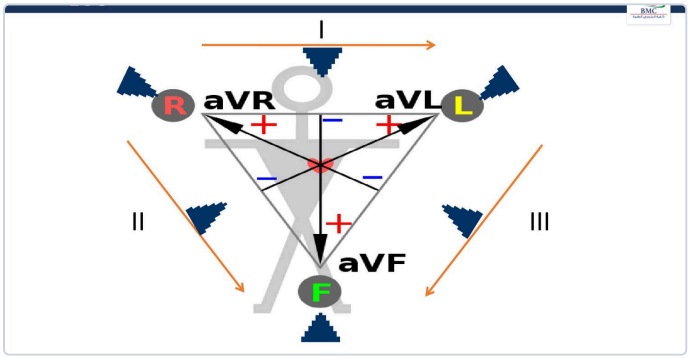
- **Placement (slide 11):** - **V1** 4th ICS right sternal edge; **V2** 4th ICS left - **V3** between V2 and V4; **V4** 5th ICS **midclavicular** - **V5** anterior axillary line; **V6** midaxillary line *[added: standard]*
- **Groups (slide 12):** - **V1–V2 = septal** - **V3–V4 = anterior** - **V5–V6 = lateral**
- **Hexaxial angles (slide 14):** - **I = 0°** - **II = +60°** - **aVF = +90°** - **III = +120°** - **aVL = −30°** - **aVR = −150°**



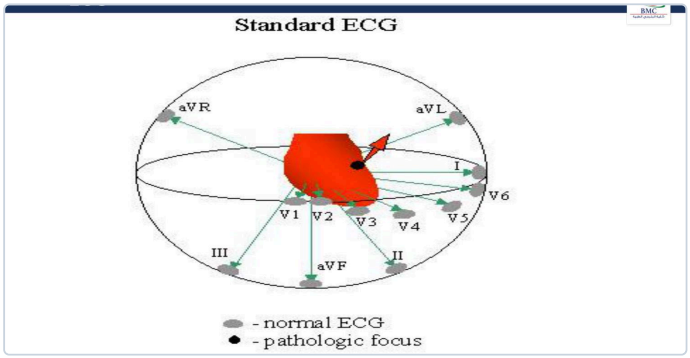
Slide 11: precordial electrode placement



Slide 12: precordial lead axes (septal / anterior / lateral)



Slide 13: Einthoven triangle and augmented limb leads



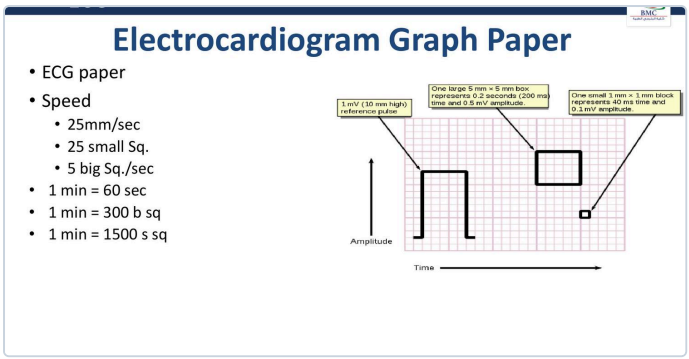
Slide 16: the 12 leads viewing the heart

2.4 Principles & Paper (Slides 17–20) ⭐

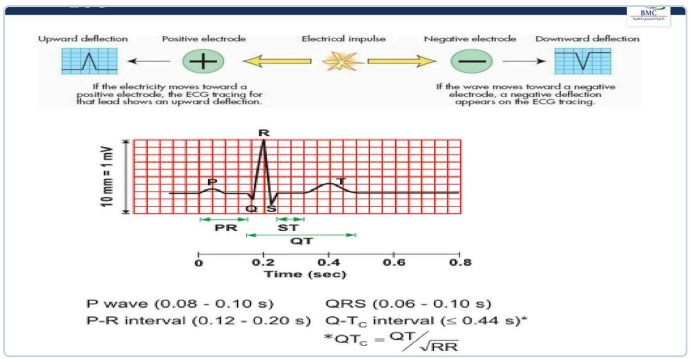
- ⭐ **Depolarisation moving toward a positive electrode → upward deflection**; moving away → downward.
- **Paper speed 25 mm/s**: - **1 small square = 0.04 s** (1 mm, 0.1 mV) - **1 big square = 0.20 s** (5 mm, 0.5 mV) - **5 big squares = 1 s**
- **1 min = 60 s = 300 big squares = 1500 small squares.**
- **Calibration:** 1 mV = 10 mm high.

Normal values (slide 19):

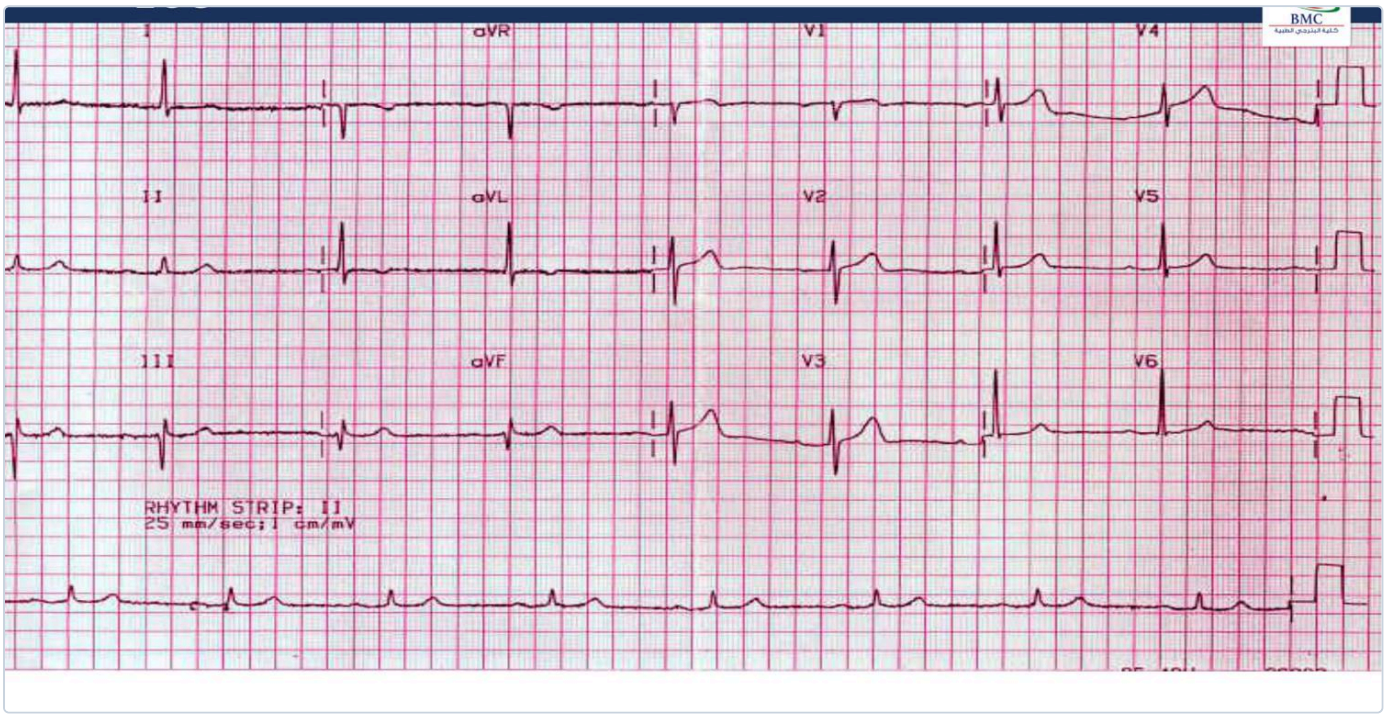
Component	Normal
P wave	**0.08–0.10 s**
PR interval	**0.12–0.20 s**
QRS	**0.06–0.10 s**
QTc	**≤ 0.44 s**; **QTc = QT/√RR** (Bazett)



Slide 17: ECG graph paper



Slide 19: deflection rule and normal intervals



Slide 20: normal 12-lead ECG

🔁 **Recap: Basics.** SA 60–100 → AV 40–60 → ventricles 20–40 (fail-safe). 6 limb leads (I, II, III bipolar; aVR, aVL, aVF unipolar) + V1–V6. Toward + = up. 25 mm/s: small square 0.04 s, big square 0.2 s, 300 big squares/min. Normal: P ≤ 0.10, PR 0.12–0.20, QRS ≤ 0.10, QTc ≤ 0.44.

🔗 **Link:** With the grid and the leads understood, every ECG is read in the same order.

▶️ **Up next: the 11-step comment.**

2.5 The 11-Step "ECG Comment" (Slide 21) ⭐

1. Identification → 2. Standardisation → 3. Voltage → 4. Axis → 5. Rhythm → 6. Rate → 7. P wave → 8. PR interval → 9. QRS → 10. ST segment → 11. T wave

(Slides 33, 67, 83 and 90 swap 5 and 6 to "rate, rhythm". The content is the same.)

Mnemonic *[added]*: **"I Saw Very Angry Rabbits Running; Puppies Played Quietly, Sleeping Together"**.

The lecturer's 11-step ECG comment, with the numbers to check

1. Identification
Name, age, date/time, clinical context

2. Standardisation
Calibration box 10 mm = 1 mV (2 big squares) · 25 mm/s

3. Voltage
High: (S V1 or V2) + (R V5 or V6) ≥ 35 mm → LVH
Low*: QRS < 5 mm in all limb leads (< 10 mm chest)

4. Axis
Normal −30° to +90° · LAD −30 to −90 · RAD +90 to +180
Quick: lead I + lead II both upright = normal

5. Rhythm
Regular vs irregular: compare R–R with calipers/paper

6. Rate
Regular: 300 ÷ big squares, or 1500 ÷ small squares (R–R)
Irregular: QRS in 6 s (30 big squares) × 10
Triplicate: 300–150–100–75–60–50

7. P wave
Upright I & II, inverted aVR · height ≤ 2.5 mm · width ≤ 0.10 s
Tall (> 2.5 mm) = P pulmonale (RA) · wide/notched = P mitrale (LA)
Absent: AF, flutter, junctional, SVT

8. PR interval
0.12–0.20 s (3–5 small squares)
Long → AV block · Short → pre-excitation (WPW), junctional

9. QRS
Normal ≤ 0.08–0.10 s · ≥ 0.12 s = bundle branch block / ventricular
RBBB: rSR′ "M" in V1, wide S in V6 · LBBB: deep QS V1, "M" V6
Pathological Q: > 1 small sq wide or > ¼ R height

10. ST segment
Iso-electric · ↑ in ≥ 2 contiguous leads = STEMI (territory)
↓ = ischaemia / reciprocal / NSTEMI · diffuse ↑ + PR↓ = pericarditis

11. T wave
Normally inverted aVR (± V1, III) · peaked = ↑K⁺ · deep ↓ = ischaemia

STEMI territory map (leads → artery)

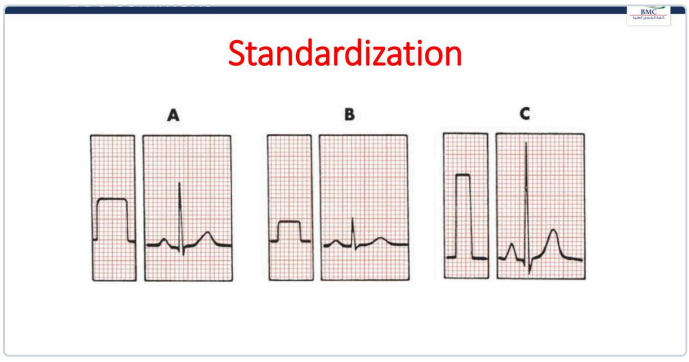
Inferior II, III, aVF → RCA (reciprocal I, aVL) · add V4R for RV infarct
Septal V1–V2 · **Anterior** V3–V4 → LAD · **Lateral** I, aVL, V5–V6 → LCx/diagonal
Posterior ST↓ + tall R in V1–V3 → confirm ST↑ in V7–V9 · **Extensive anterior** V1–V6 + I, aVL → proximal LAD

Paper: 1 small square = 1 mm = 0.04 s = 0.1 mV · 1 big square = 5 mm = 0.20 s · 5 big squares = 1 s · 300 big = 1 min
* low-voltage cut-offs and the territory map are added (slides give only the headings); everything else is from the slides.

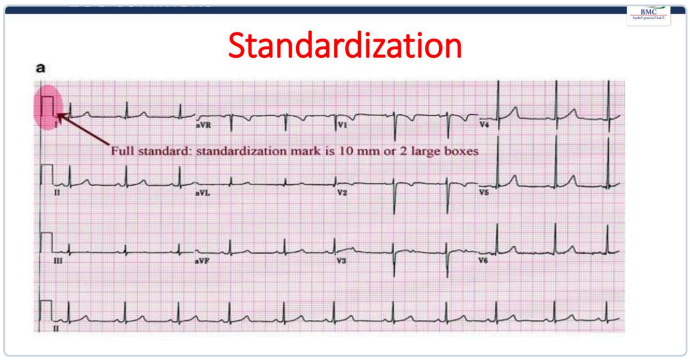
Added diagram: the 11 steps with normal values and the STEMI territory map

2.6 Steps 1–2: Identification & Standardisation (Slides 23–25)

- **Identification:** patient name, age, date and time, clinical context *[added detail]*.
- **Standardisation:** the calibration box at the start of each row. - ⭐ **Full standard = 10 mm (2 big boxes)** tall (slide 24). - Slide 23: **A = full standard (10 mm)** · **B = half standard (5 mm**; complexes look half-size) · **C = double standard (20 mm**; complexes look twice as big).
- ⭐ **Always check calibration before judging voltage.** A half-standard trace mimics low voltage; double standard mimics LVH.



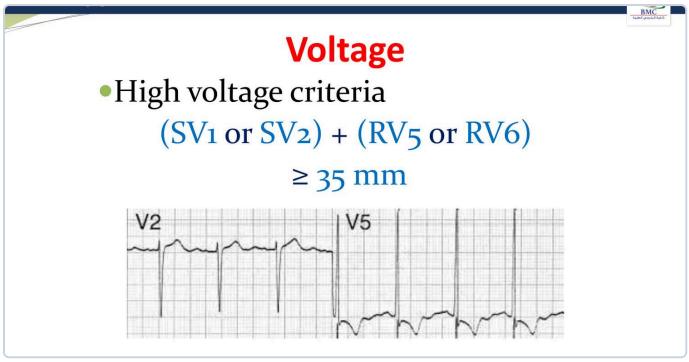
Slide 23: full vs half vs double standard



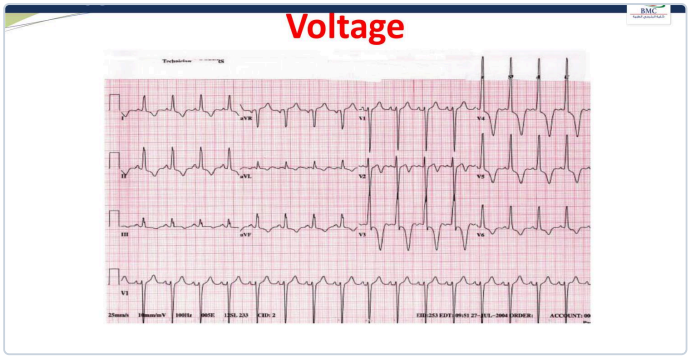
Slide 24: full standard (10 mm)

2.7 Step 3: Voltage (Slides 27–32) ⭐

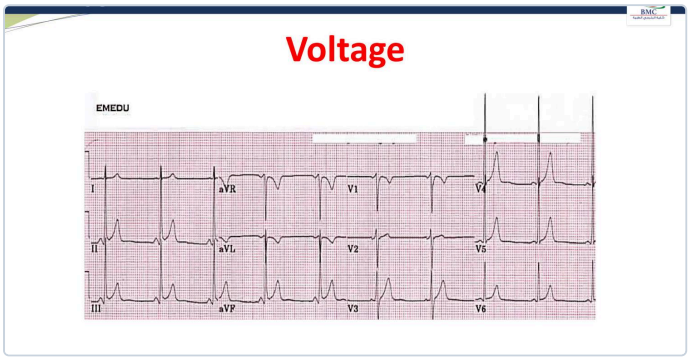
- **High voltage:** ⭐ **(S in V1 or V2) + (R in V5 or V6) ≥ 35 mm** = **LVH** (Sokolow–Lyon).
- **Low voltage** *[added: criteria not on slides]*: QRS **< 5 mm in all limb leads** or **< 10 mm in all chest leads**. Causes: **pericardial effusion**, obesity, COPD/emphysema, hypothyroidism, infiltration (amyloid), pleural effusion.
- **Normal voltage**.



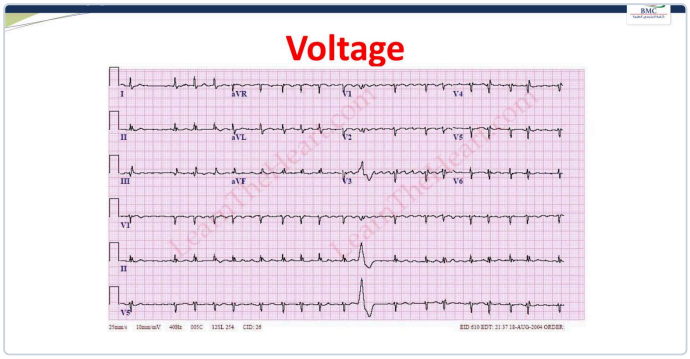
Slide 28: high-voltage criteria (V2 and V5)



Slide 32: high voltage, LVH (🔎 my reading: tall R in V4–V6, deep S in V1–V3)



Slide 30: practice voltage ECG



Slide 31: low voltage (🔎 my reading: small complexes in every lead; isolated wide ectopic)

2.8 Step 4: Axis (Slides 34–41) ⭐

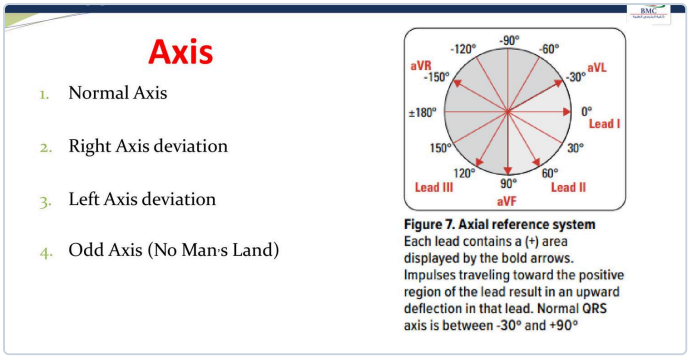
Categories: normal · right axis deviation · left axis deviation · **odd axis ("no man's land")**.

Axis	Range	Lead I	Lead II	Lead III / aVF
⭐ **Normal**	**0° to +90°** (slide 35 figure: **−30° to +90°**)	↑	↑	↑
Left, physiological	**0° to −30°**	↑	↑ / biphasic	↓
⭐ **Left, pathological**	**−30° to −90°**	↑	**↓**	↓
⭐ **Right**	**+90° to +180°**	**↓**	↑	↑
Extreme ("no man's land")	**−90° to ±180°**	↓	↓	↓
Indeterminate	?	biphasic	biphasic	biphasic

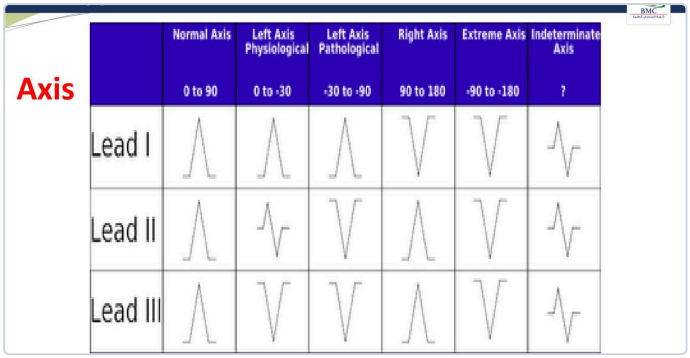
> **Quick rule** *[added]*: **"Leaving" (left)** = leads I and aVF point **away** from each other (I up, aVF down). **"Reaching" (right)** = I and aVF point **toward** each other (I down, aVF up). Then use **lead II** to split physiological from pathological left axis.

[added]:

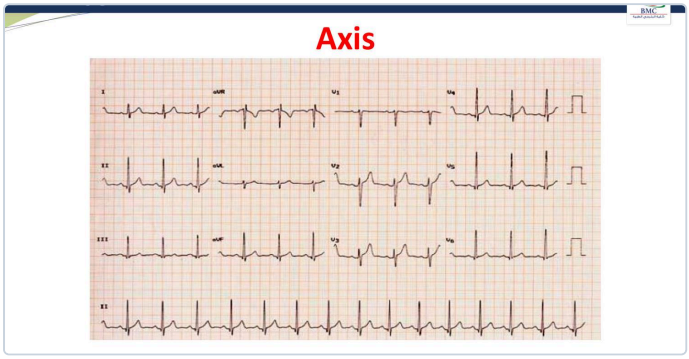
- **LAD causes: left anterior fascicular block**, inferior MI, LVH, WPW.
- **RAD causes: RVH** (pulmonary HTN, PE, COPD), lateral MI, left posterior fascicular block, dextrocardia, normal in children and thin adults.



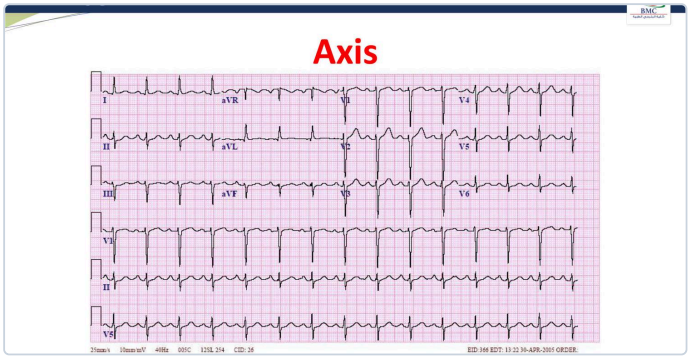
Slide 35: hexaxial reference system



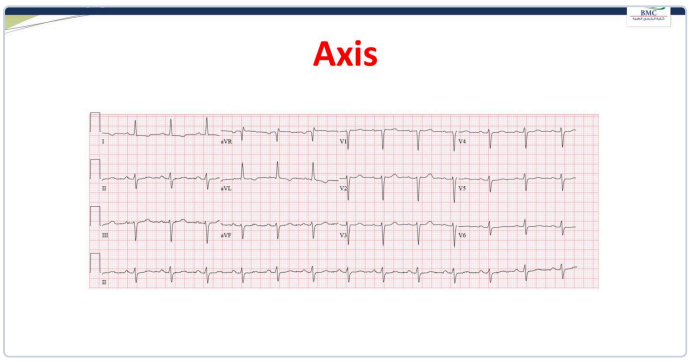
Slide 36: axis by leads I, II, III



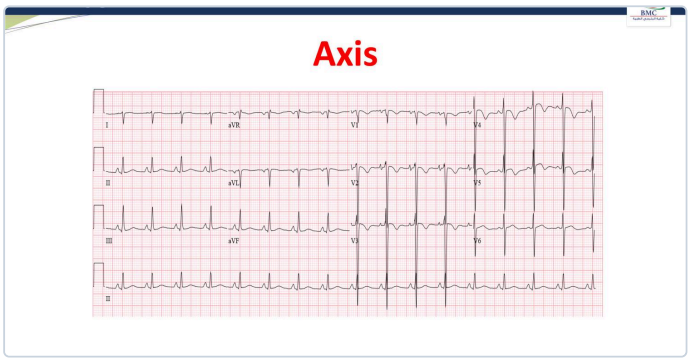
Slide 37: axis practice (🔎 normal axis: I, II, III all positive)



Slide 39: axis practice (🔎 left axis: I positive, II and aVF negative)



Slide 40: axis practice



Slide 41: axis practice

> 🔁 **Recap: Steps 1–4.** Identify the patient and context. Check calibration (10 mm = full standard). Voltage: S V1/V2 + R V5/V6 ≥ 35 → LVH; small complexes → effusion, obesity, COPD, hypothyroid. Axis: normal −30 to +90 (I and II up); LAD = I up, aVF and II down; RAD = I down, aVF up.
>
> 🔗 **Link:** Next is the timing of the beats: regular or not, and how fast.
>
> ▶️ **Up next: Rhythm and rate.**

2.9 Step 5: Rhythm (Slides 43–55) ⭐

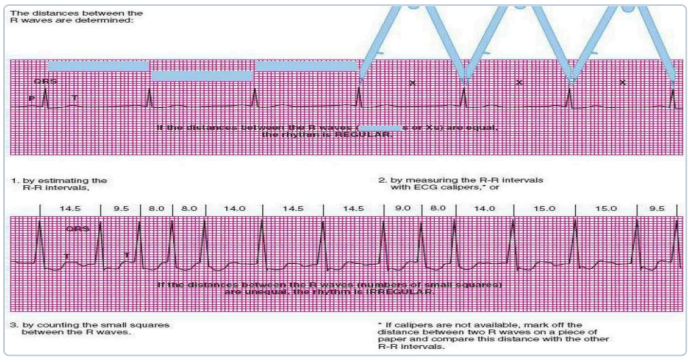
Regular vs irregular (slide 44). Measure the **R–R intervals**:

1. by eye
2. with **calipers**
3. by **counting small squares** between R waves

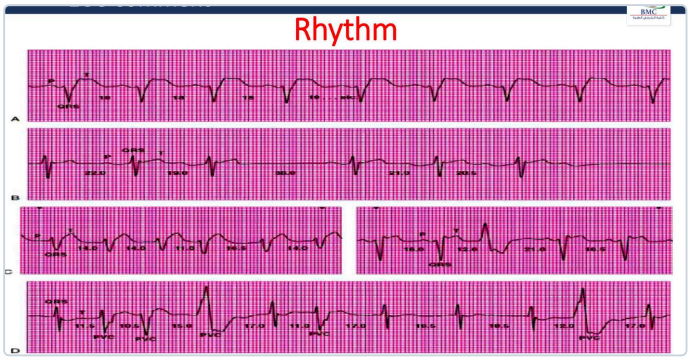
Equal intervals mean the rhythm is regular.

Slide 45 (irregular patterns):

Strip	🔎 Reading
A	Regular
B	Irregular (varying R–R: 22, 19, 35, 21, 20 small squares)
C	Occasional irregularity from premature beats
D	Irregular with **PVCs** (labelled)



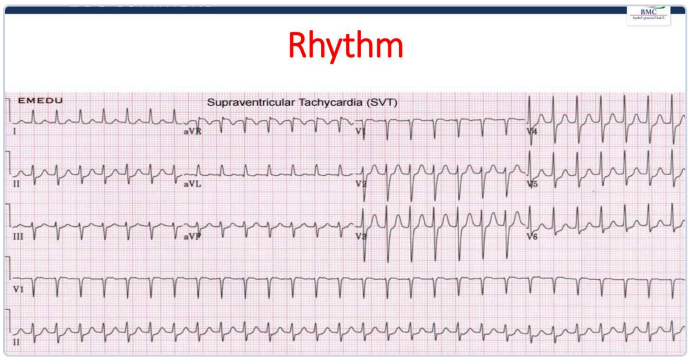
Slide 44: measuring R–R intervals



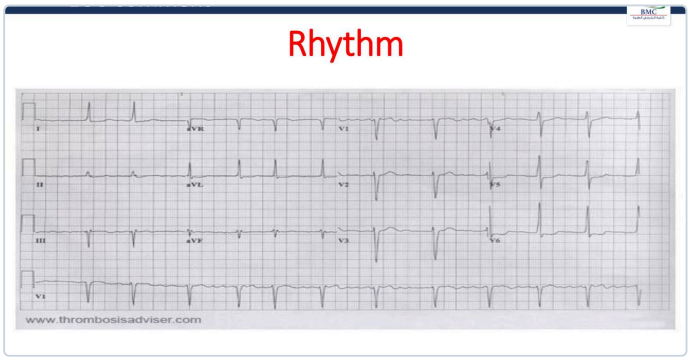
Slide 45: regular vs irregular strips (A–D)

Rhythm practice strips:

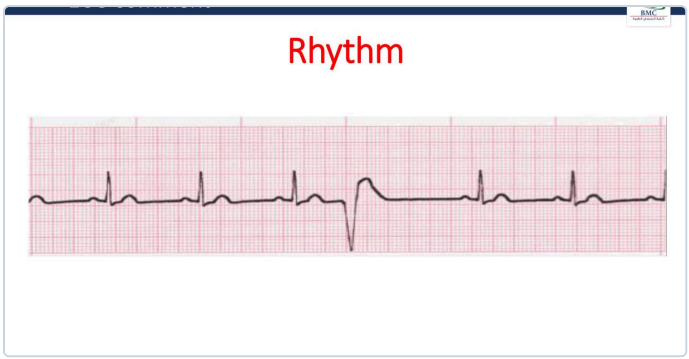
Slide	🔎 Reading
47	**SVT** (labelled): regular narrow-complex tachycardia, no visible P
48	Irregularly irregular narrow complexes, no clear P waves → **AF**
49	Sinus rhythm with **one PVC** (wide, bizarre, early, compensatory pause)
50	Irregular, fibrillatory baseline in V1 → **AF (controlled rate)**
51	Irregularly irregular narrow complexes on rhythm strips → **AF**
52	Sinus with **frequent PVCs** (multifocal wide beats)
53	Sinus with **a single PVC**



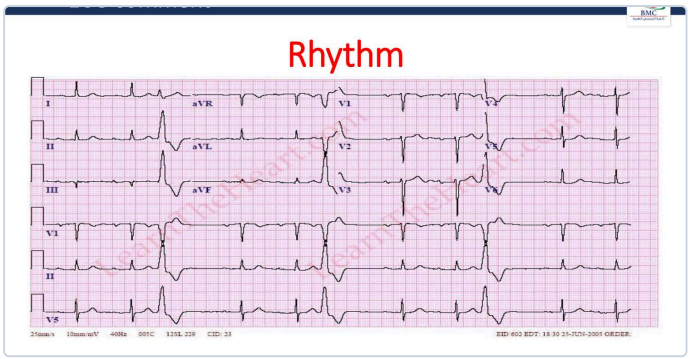
Slide 47: SVT



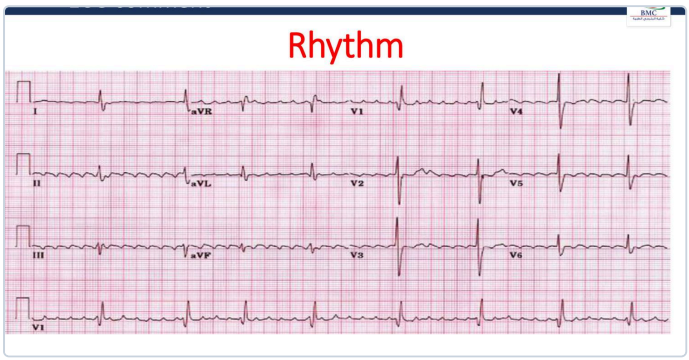
Slide 48: irregularly irregular rhythm



Slide 49: isolated PVC



Slide 52: frequent PVCs



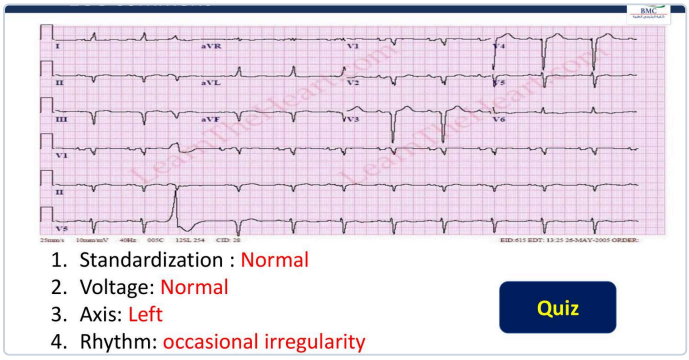
Slide 50: AF, 12-lead



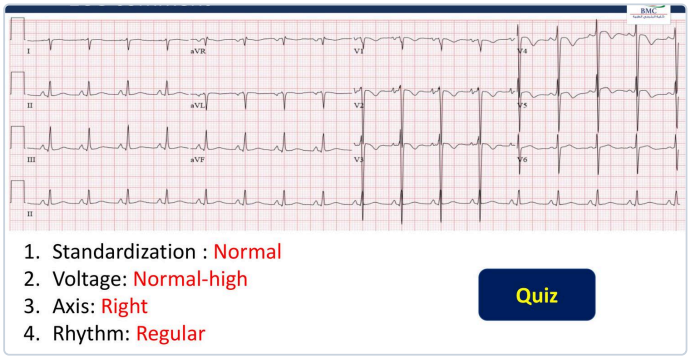
Slide 51: irregular rhythm strips

Lecturer's quiz answers (slides 54–55):

Slide	Standardisation	Voltage	Axis	Rhythm
54	Normal	Normal	**Left**	**Occasional irregularity** (🔎 *ectopic beats*)
55	Normal	**Normal–high**	**Right**	**Regular** (🔎 *consistent with RVH*)



Slide 54: quiz (left axis, occasional irregularity)

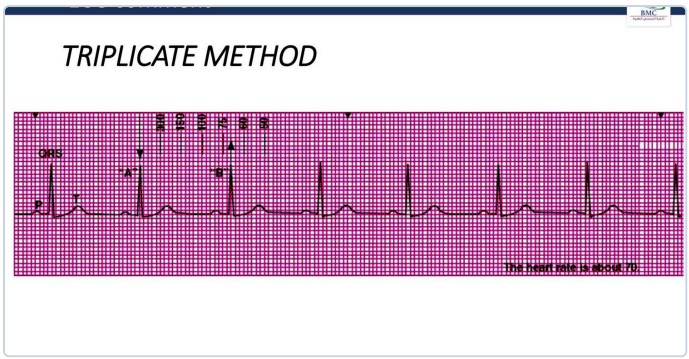


Slide 55: quiz (right axis, normal–high voltage)

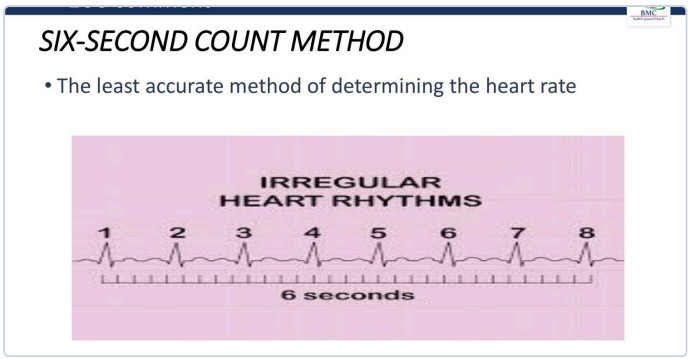
2.10 Step 6: Rate (Slides 57–65) ⭐

Method	Use	Formula
Heart-rate ruler (slide 57)	Any	Read the scale at the 3rd complex
⭐ **Triplicate / sequence method** (slide 58)	**Regular**	Start at an R on a heavy line; count the next heavy lines **300 – 150 – 100 – 75 – 60 – 50** (example ≈ **70**)
⭐ **R-R method** (slides 59–60)	**Regular**	**HR = 300 ÷ big squares** between R waves · **HR = 1500 ÷ small squares**
⭐ **6-second count** (slides 61–62)	**Irregular**	**QRS in 6 s (= 30 big squares = 1/10 min) × 10**. ⭐ **The least accurate method**

Slide 59 logic: rate = QRS in 1 min × 1 = QRS in ½ min × 2 = … = **QRS in 1/10 min × 10**.



Slide 58: triplicate method



Slide 61: 6-second count method

Mnemonic: "300-150-100, 75-60-50."

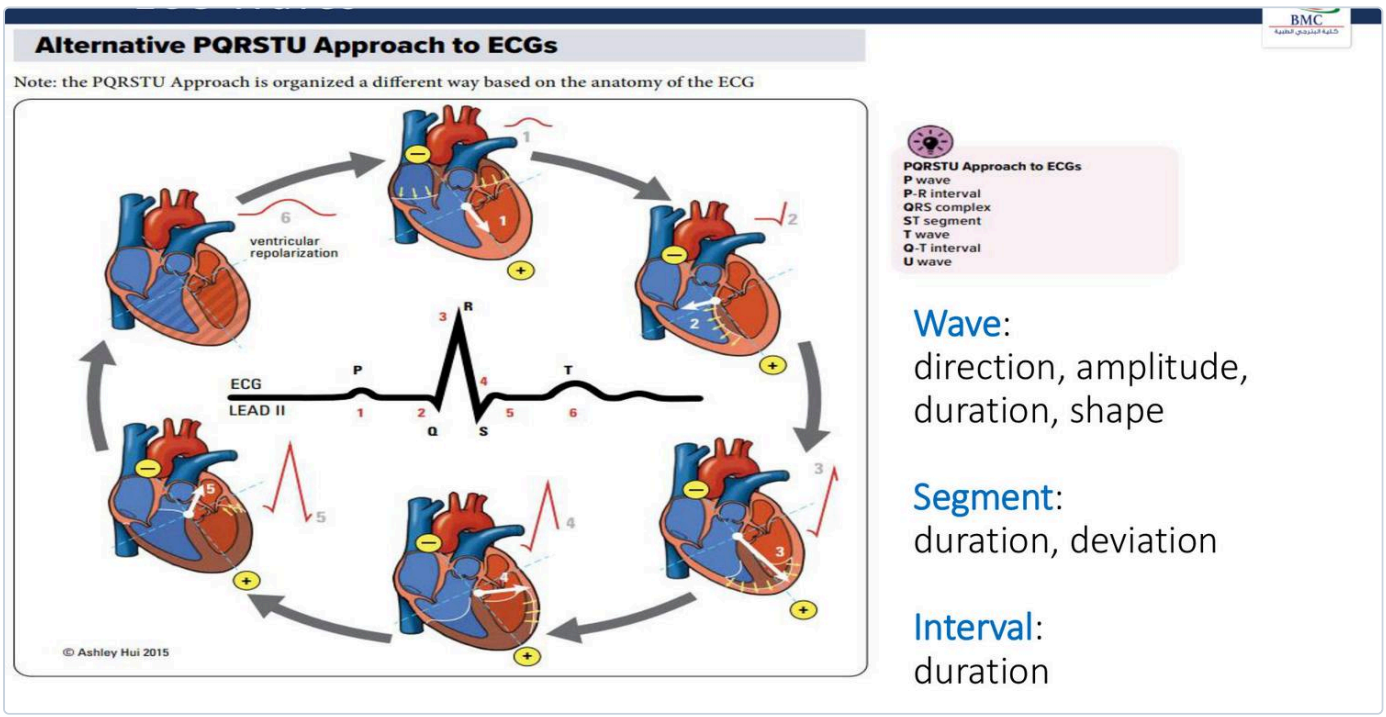
🔁 **Recap: Steps 5–6.** Rhythm: compare R–R intervals (eye, calipers, small squares). Rate: regular → 300/big squares or 1500/small squares (or the 300-150-100 sequence); irregular → beats in 6 s × 10 (least accurate).

🔗 **Link:** Then go through each wave and interval in order of the cardiac cycle, starting with atrial depolarisation.

▶️ **Up next: Waves and intervals.**

2.11 Waves, Segments, Intervals (Slide 66)

- **Wave:** direction, amplitude, duration, shape.
- **Segment:** duration, deviation.
- **Interval:** duration.



Slide 66: waves, segments, intervals

2.12 Step 7: P Wave (Slides 68–82) ⭐

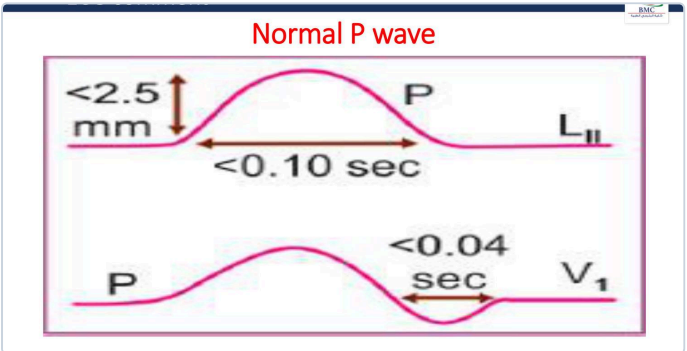
Feature	Normal
Direction	⭐ **Upright in I** (and II); **inverted in aVR**

Feature	Normal
Amplitude (right atrium)	**≤ 2.5 mm**
Duration (left atrium)	**≤ 2.5 mm = 0.10 s**
Shape	Best seen in **II and V1** (biphasic in V1; terminal negative part < 0.04 s)

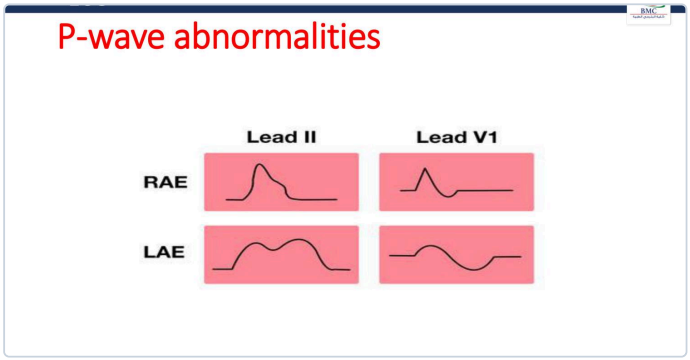
Analyse the P waves (slide 69):

1. Present?
2. Regular?
3. One P per QRS, and a QRS after every P?
4. Upright or inverted?
5. All alike? (Alike and regular means the same origin.)

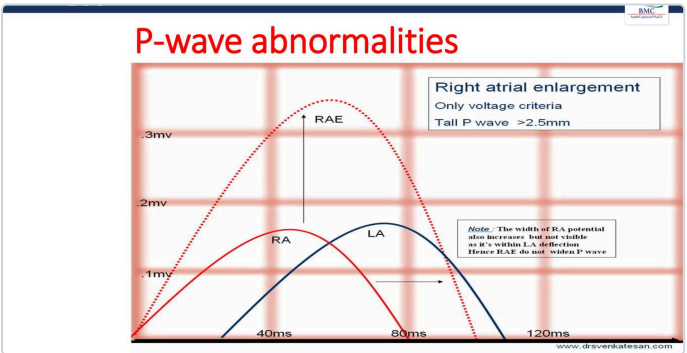
⭐ Abnormality	Criterion	Meaning
Abnormal direction	Not upright in I / not inverted in aVR	**Ectopic atrial / junctional rhythm**; *[added]* limb-lead reversal, dextrocardia
⭐ **P pulmonale**	**Height > 2.5 mm** (II)	**Right atrial enlargement** (*[added]* pulmonary HTN, COPD, TR)
⭐ **P mitrale**	**Width > 0.10 s (> 2.5 mm)**, **notched**, notch > 40 ms; **P terminal force in V1 > 0.04 s**	**Left atrial enlargement** (*[added]* **mitral stenosis**, MR, LVH)
Absent / replaced	**Fibrillatory** (AF), **flutter** waves, **nodal** rhythm, **SVT**	Non-sinus rhythm



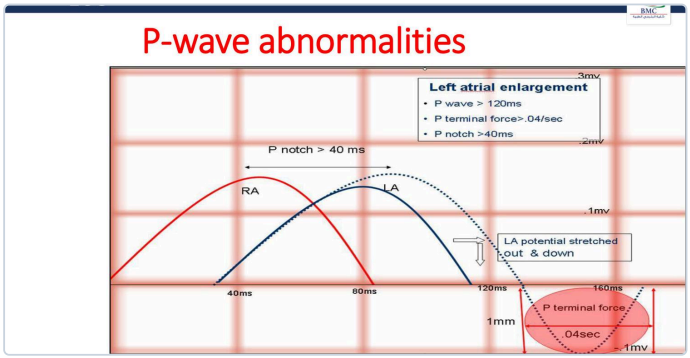
Slide 71: normal P wave (< 2.5 mm, < 0.10 s; V1 negative part < 0.04 s)



Slide 75: RAE vs LAE in leads II and V1



Slide 72: right atrial enlargement (tall P)

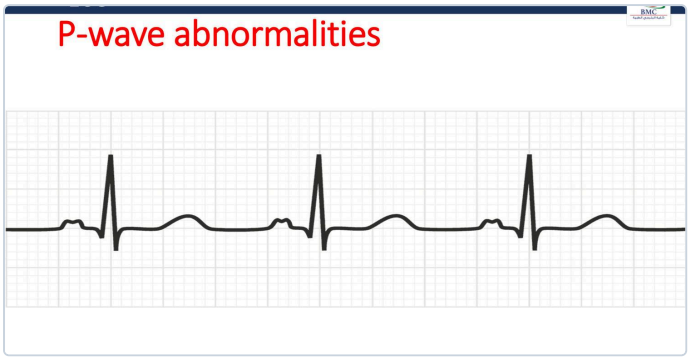


Slide 73: left atrial enlargement (wide, notched P)

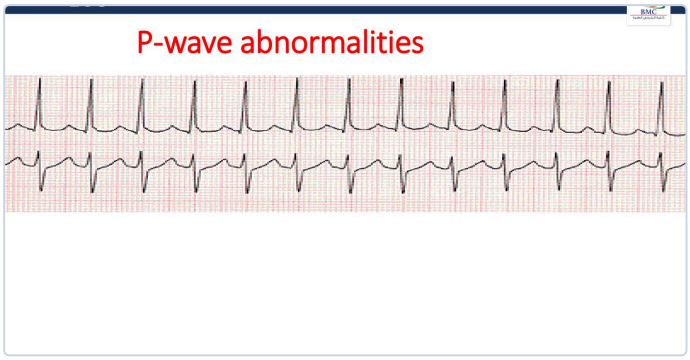
P-wave practice strips:

Slide	🔎 Reading
77	Notched, bifid P waves → **P mitrale (LAE)**
78	Tall, peaked P → **P pulmonale (RAE)**
79	**SVT** (labelled): no visible P

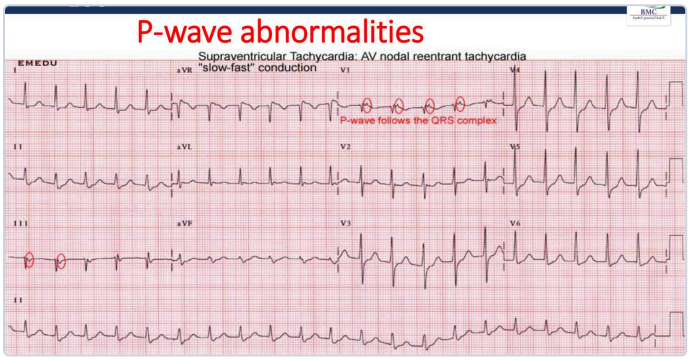
Slide	🔎 Reading
80	**AVNRT** (labelled "slow-fast"): **P wave follows the QRS** (retrograde)
81	P waves positioned around/after the QRS → junctional / retrograde P
82	**Atrial flutter** (labelled): sawtooth flutter waves on the V1 rhythm strip



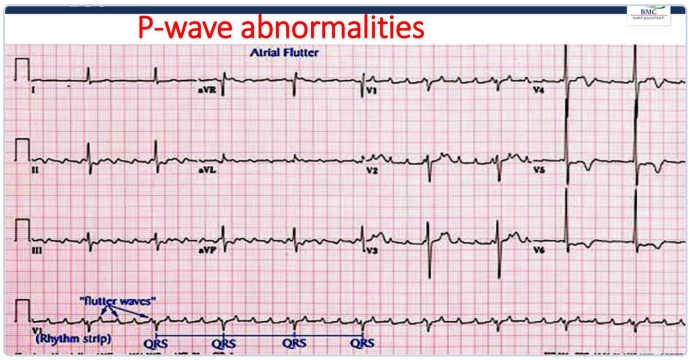
Slide 77: notched P (P mitrale)



Slide 78: tall P (P pulmonale)



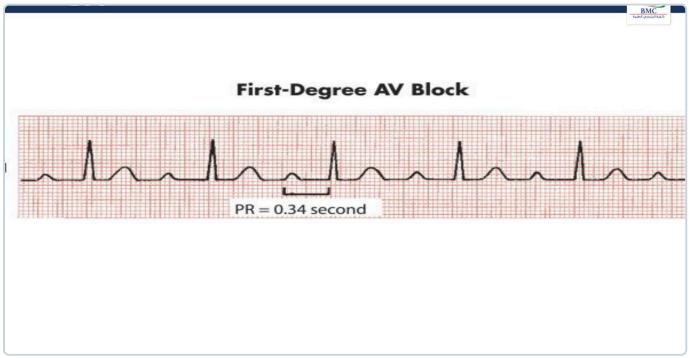
Slide 80: AVNRT, P after QRS



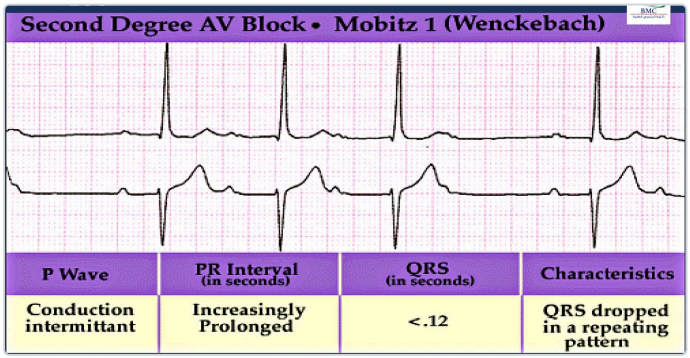
Slide 82: atrial flutter, flutter waves

2.13 Step 8: PR Interval (Slides 84–89) ⭐

- **Definition** *[added]*: from the start of P to the start of QRS. **Represents** conduction from the atria through the **AV node** and His–Purkinje system.
- ⭐ **Normal: 3–5 small squares = 0.12–0.20 s.**
- **Prolonged → AV block**: - **1st degree**: PR > 0.20 s, every P conducted. Slide 86: **PR = 0.34 s**. - **2nd degree Mobitz I (Wenckebach)** (slide 87): conduction **intermittent**; PR **increasingly prolonged** until a QRS is dropped in a **repeating pattern**; QRS **< 0.12 s**. - *[added]* **Mobitz II**: constant PR, sudden dropped QRS. **3rd degree**: AV dissociation.
- **Shortened** *[added]*: **pre-excitation (WPW: short PR + delta wave)**, LGL, junctional rhythm.



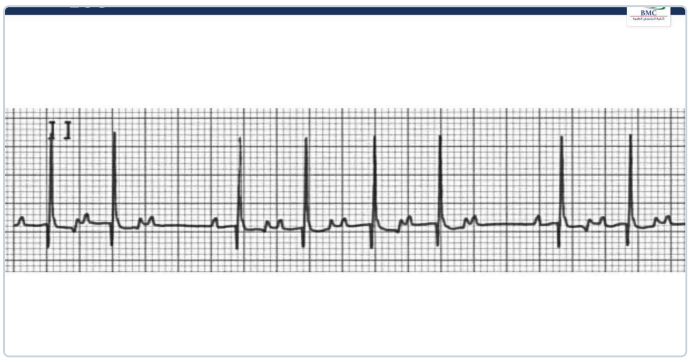
Slide 86: first-degree AV block (PR 0.34 s)



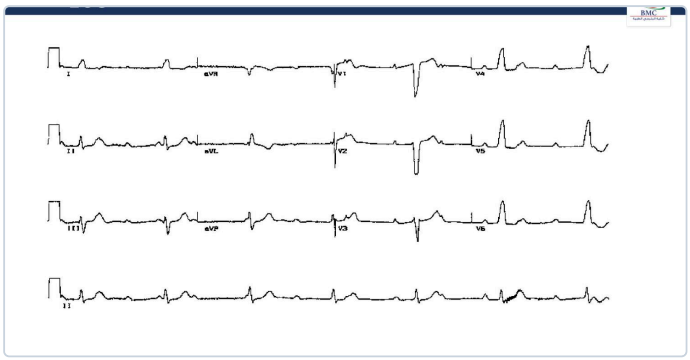
Slide 87: Mobitz I (Wenckebach)

PR practice strips:

Slide	🔎 Reading
85	12-lead for PR assessment (🔎 *short PR; check for a delta wave*)
88	Lead II: **more P waves than QRS**, some P waves not conducted → **2nd-degree AV block** (🔎 *constant PR on conducted beats suggests Mobitz II/2:1*)
89	Slow rate, P waves marching through independently of the QRS → **complete (3rd-degree) heart block**



Slide 88: P waves not all conducted



Slide 89: AV dissociation

🔁 **Recap: P and PR.** P upright in I/II, inverted in aVR; ≤ 2.5 mm tall and ≤ 0.10 s wide. Tall = RA (pulmonale); wide/notched = LA (mitrale). Absent = AF/flutter/junctional/SVT; retrograde after QRS = AVNRT. PR 0.12–0.20: long = AV block (1st 0.34 s; Wenckebach progressive); short = WPW/junctional.

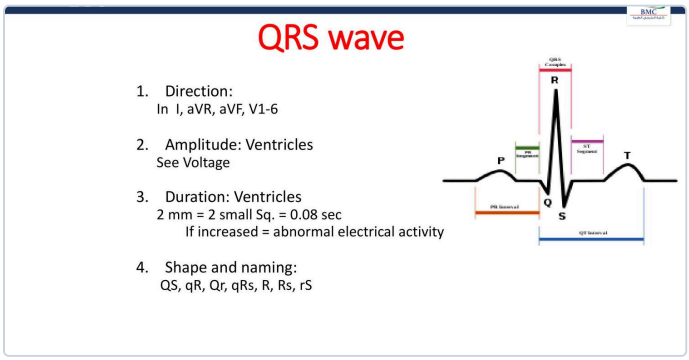
🔗 **Link:** After the AV node, conduction runs down the bundle branches, so the QRS tells you whether that is normal.

▶️ **Up next: QRS.**

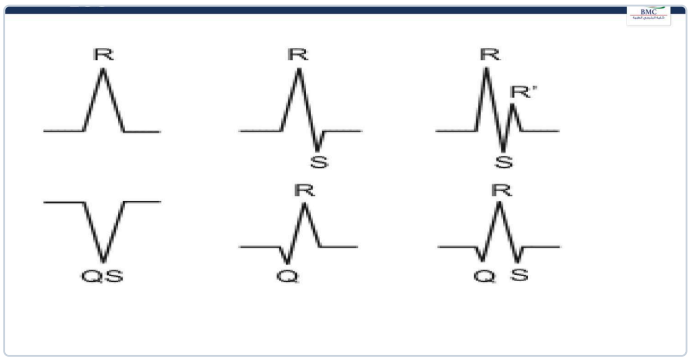
2.14 Step 9: QRS Complex (Slides 91–104) ⭐

Feature	Normal
Direction	Positive in **I, aVF, V5–V6**; **negative in aVR**; **V1 mostly negative (rS)**. R wave grows V1→V6 (**R-wave progression**)
Amplitude	See voltage
⭐ **Duration**	**≤ 2 small squares = 0.08 s** (slide 19: 0.06–0.10 s). **Increased = abnormal ventricular conduction**
Naming	**QS, qR, Qr, qRs, R, Rs, rS** (capital = big, small = small; R′ = second positive wave)

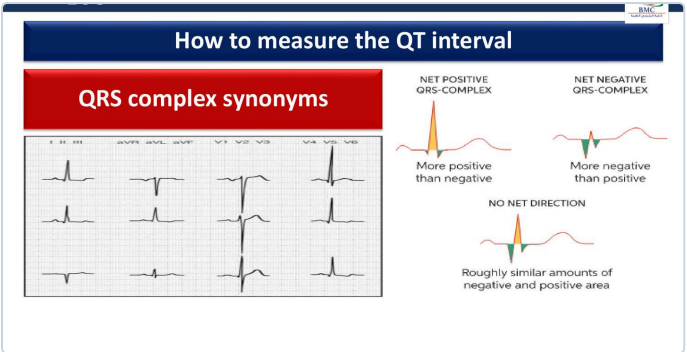
- **Rules (slides 93–94):** - **R** = any upward deflection; **Q** = negative **before** R; **S** = negative **after** R; **QS** = entirely negative. - **Net positive vs net negative** complexes decide the axis.
- **Slide 95, Quiz 5:** name 10 complexes using QS/qR/Rs/rS/RSR′…



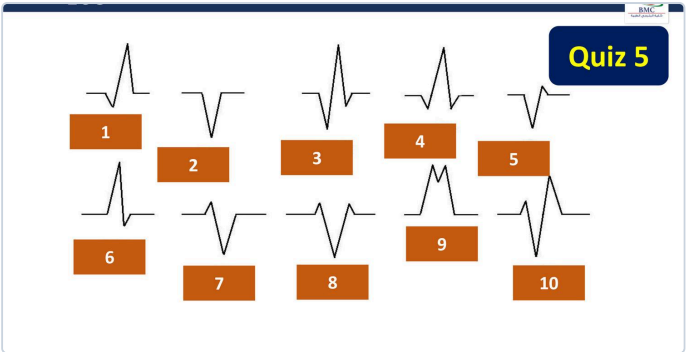
Slide 91: QRS features



Slide 94: QRS naming (R, RS, RSR′, QS, QR, QRS)



Slide 93: QRS synonyms, net positive vs net negative



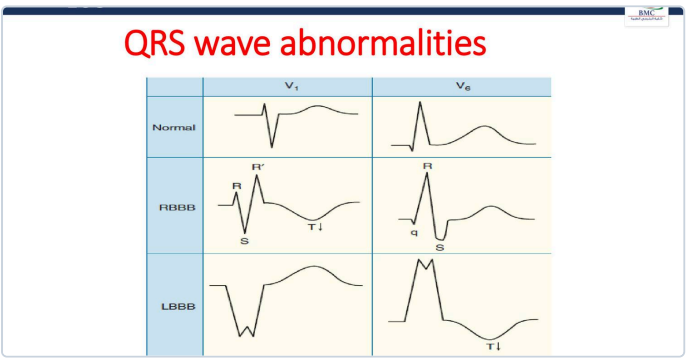
Slide 95: Quiz 5, name the complexes

Bundle branch blocks (slides 97–100) ⭐

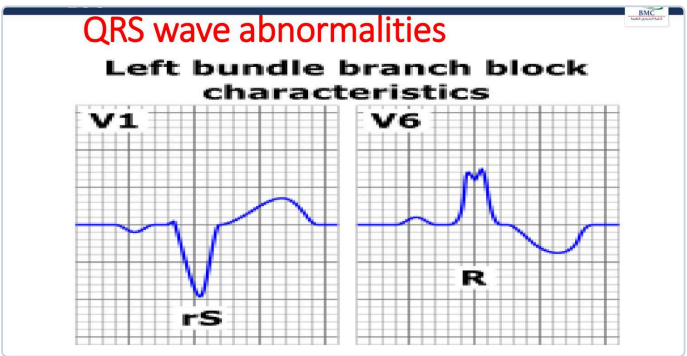
	V1	V6
Normal	rS	qR
⭐ **RBBB**	**rSR′ ("M" / rabbit ears)**, T inverted	**qRS with wide, slurred S**
⭐ **LBBB**	**Deep, wide QS / rS** ("W")	**Broad, notched R ("M")**, T inverted, no q

- **QRS ≥ 0.12 s** in both (slide 100: RBBB QRS **154 ms**).

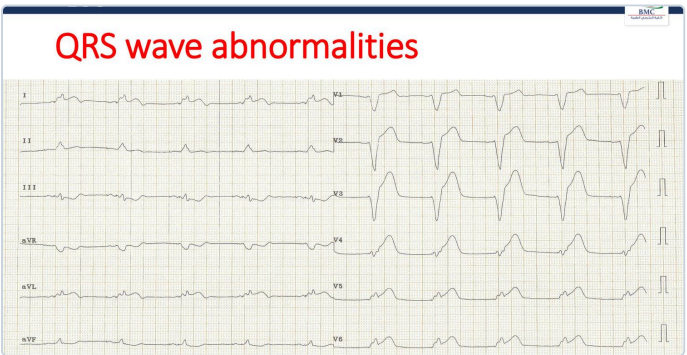
> **Mnemonic** *[added]*: **"WiLLiaM MaRRoW"**: **LBBB = W in V1, M in V6**; **RBBB = M in V1, W in V6**.



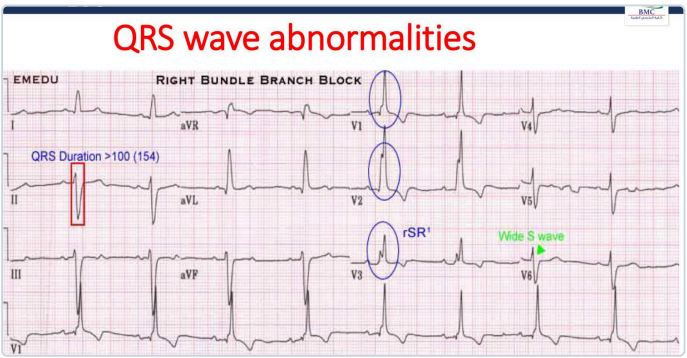
Slide 97: normal vs RBBB vs LBBB in V1 and V6



Slide 99: LBBB characteristics



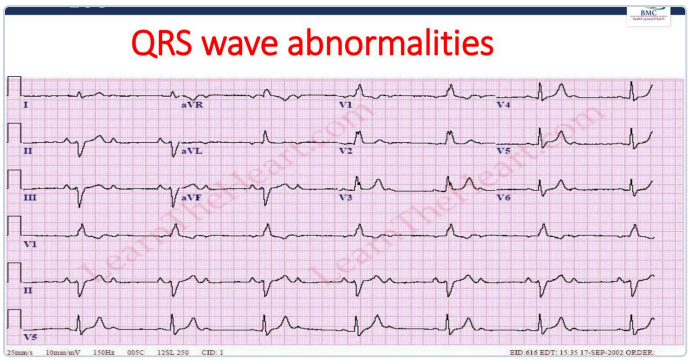
Slide 98: LBBB 12-lead



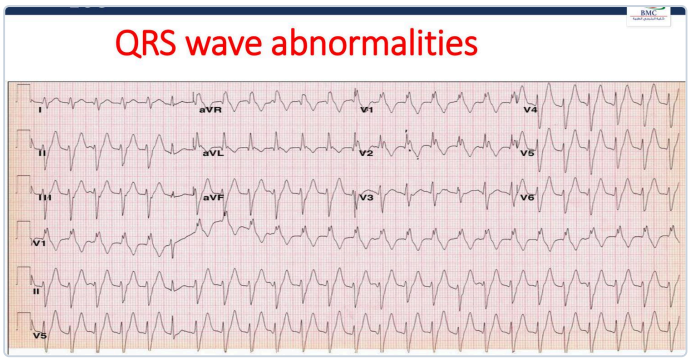
Slide 100: RBBB (rSR′ in V1–V3, wide S in V6, QRS 154 ms)

QRS practice strips:

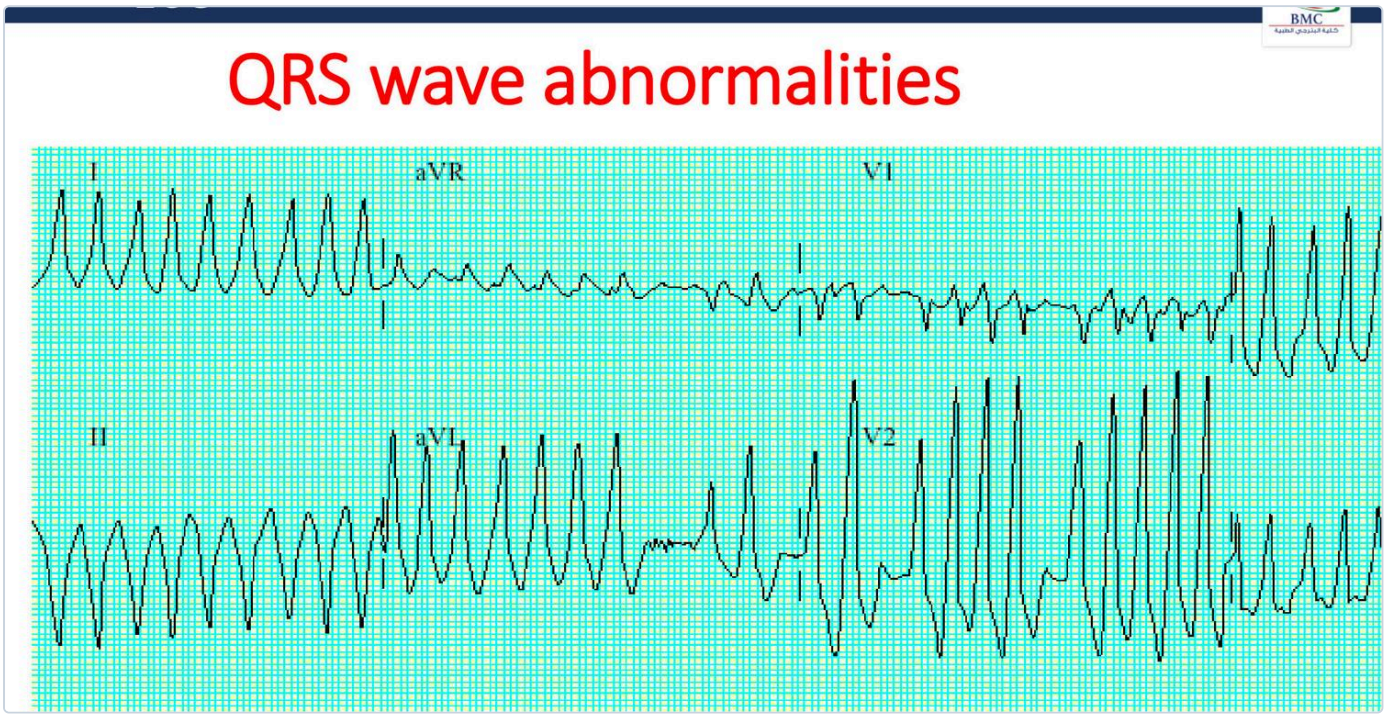
Slide	🔎 Reading
101	Deep **Q waves in III and aVF** (inferior) → **pathological Q / old inferior MI**
102	**Wide-complex tachycardia** → **VT** until proven otherwise
104	Chaotic polymorphic complexes changing amplitude → **torsades / polymorphic VT → VF**



Slide 101: inferior Q waves



Slide 102: wide-complex tachycardia



Slide 104: polymorphic VT / VF

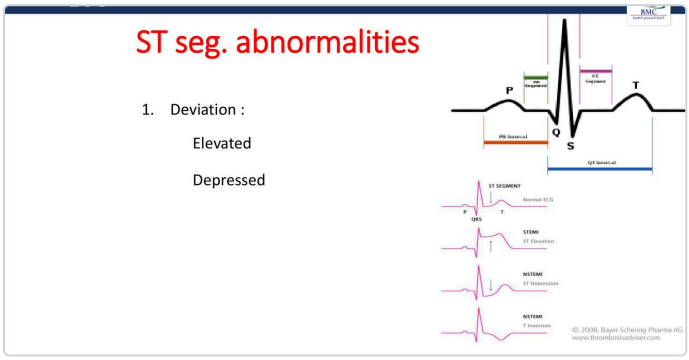
[added] Pathological Q: > 1 small square (0.04 s) wide or > ¼ of the R height, or any Q in V1–V3. It marks an old transmural MI.

2.15 Step 10: ST Segment (Slides 106–117) ⭐

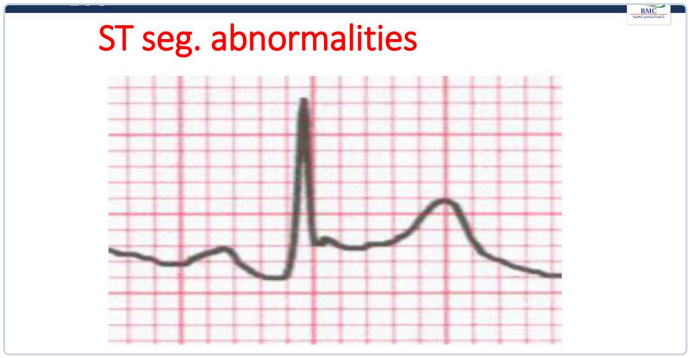
- **Definition** *[added]*: from the end of the QRS (J point) to the start of the T wave. It **represents** the plateau of ventricular depolarisation (early repolarisation).
- ⭐ **Normal: iso-electric.**
- **Deviation: elevated** (STEMI) or **depressed** (NSTEMI / ischaemia). T inversion can also be NSTEMI.

Slide	Finding
108	Single beat with **ST elevation**
⭐ **109**	**Acute INFERIOR MI**: ST↑ **3–4 mm in II, III, aVF**; **reciprocal change in I, aVL**
⭐ **110**	**Acute SEPTAL MI**: ST↑ **V2–V3**; reciprocal change in II, III, aVF
111	**Inferior ST elevation with reciprocal change** (aVL, V2, V5 labelled)
112	Inferior ST↑ with **posterior leads V7, V8, V9** recorded (handwritten) → 🔎 **infero-posterior MI**

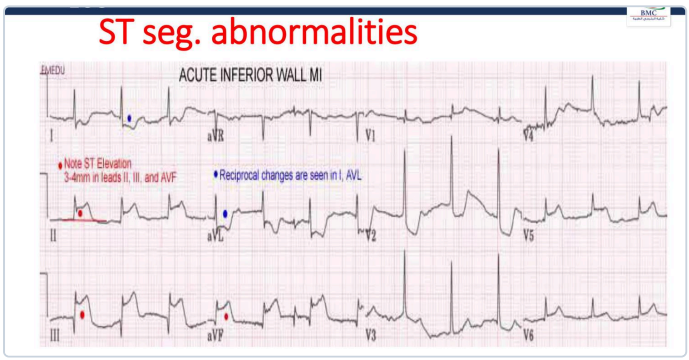
Slide	Finding
⭐ 113	**Acute inferior + right ventricular MI**: Q waves and domed ST↑ in **II, III, aVF**; **ST↑ in V4R, V5R, V6R** = RV involvement; ST↓ in I, aVL = reciprocal
114	🔎 Leads grouped I–III / aVR–aVF / V1–V3 / V4–V6 for territory practice: diffuse ST change
⭐ 115	🔎 **Extensive anterior (antero-lateral) STEMI**: tombstone ST↑ V1–V6 + I, aVL → proximal LAD
116	🔎 **Inferior STEMI** (II, III, aVF) with **reciprocal ST↓ in I, aVL** ± anterior ST↓
117	🔎 **ST depression** in lateral leads → ischaemia / **NSTEMI**



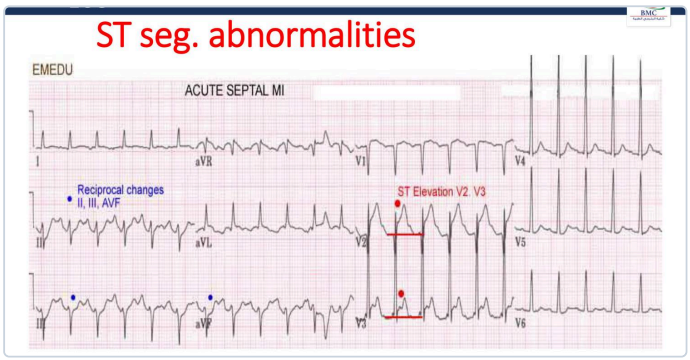
Slide 107: ST deviation types (normal, STEMI, NSTEMI ST↓, NSTEMI T inversion)



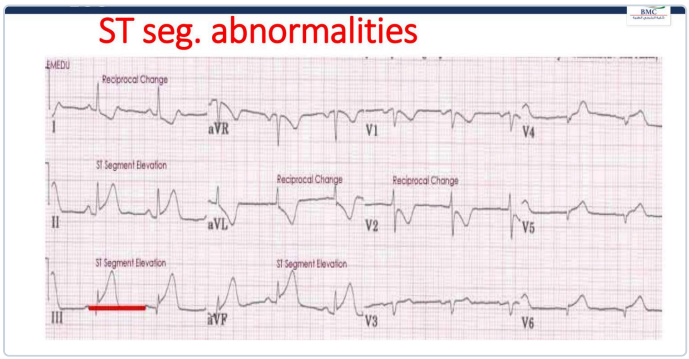
Slide 108: ST elevation



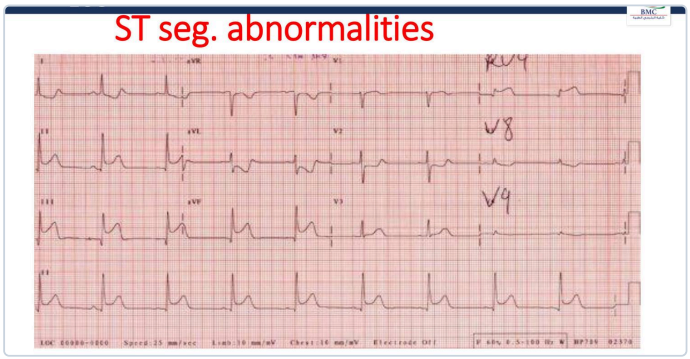
Slide 109: acute inferior MI



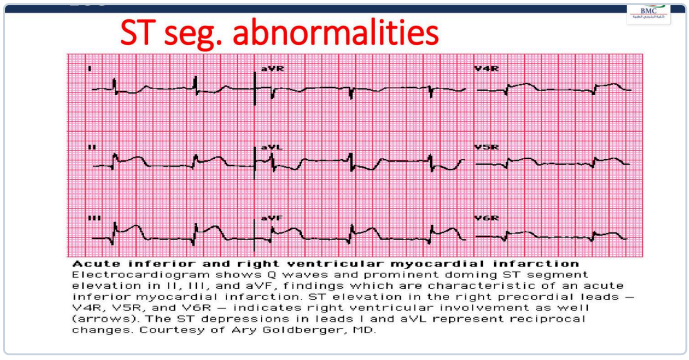
Slide 110: acute septal MI



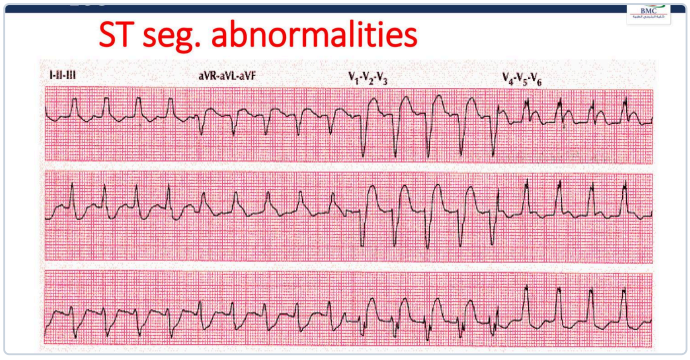
Slide 111: reciprocal change



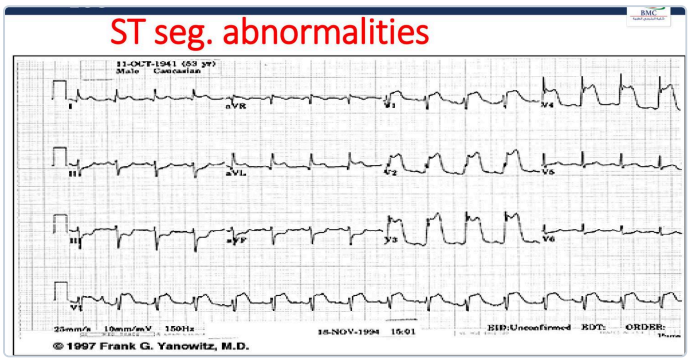
Slide 112: inferior MI with posterior leads V7–V9



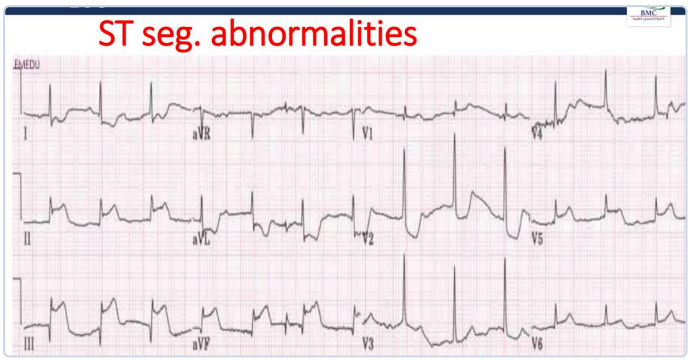
Slide 113: inferior + RV infarct (V4R–V6R)



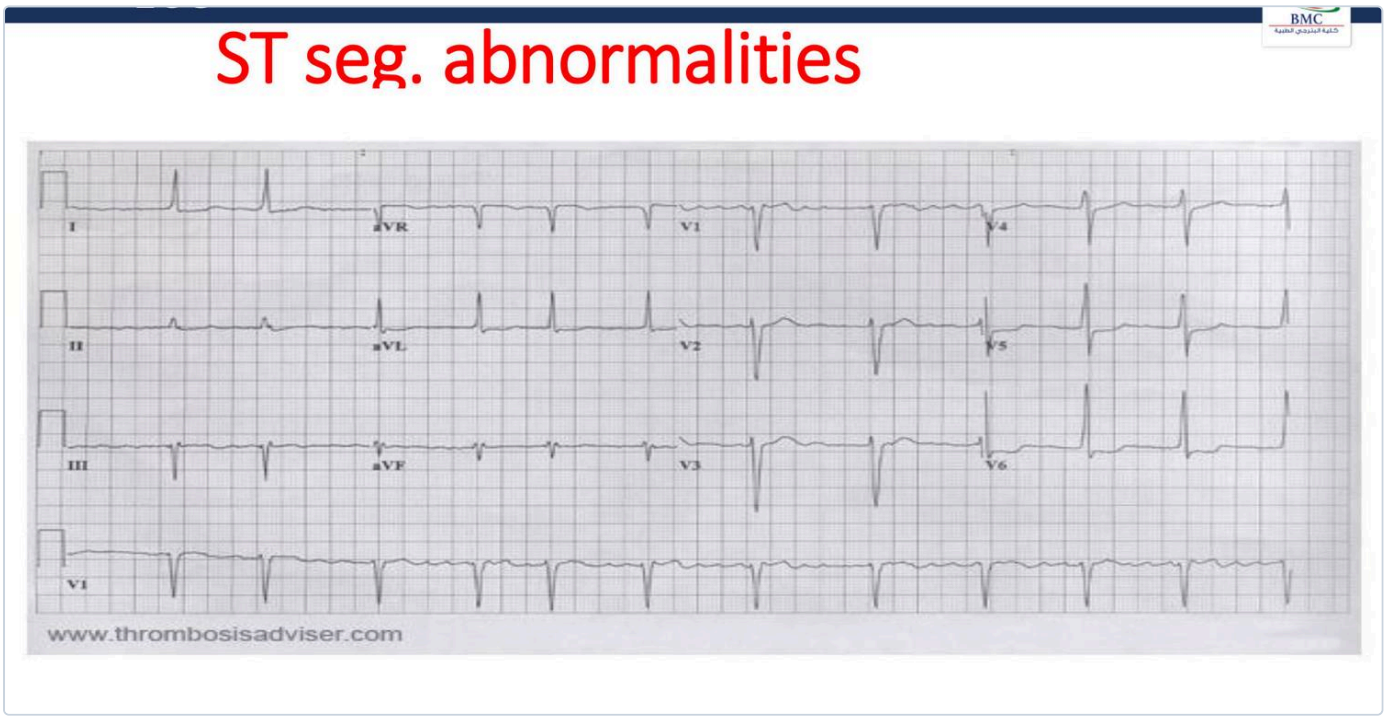
Slide 114: territory practice



Slide 115: extensive anterior STEMI



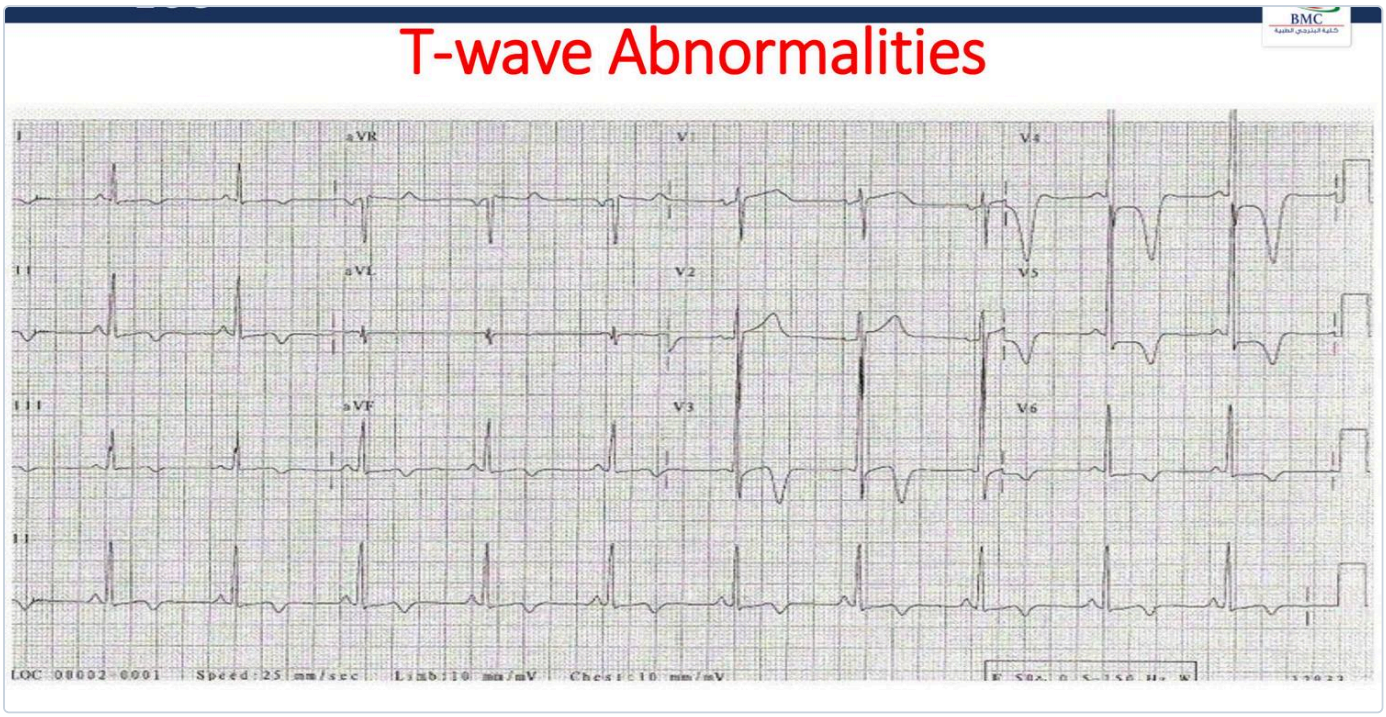
Slide 116: inferior STEMI with reciprocal change



Slide 117: ST depression

2.16 Step 11: T Wave (Slides 119–120)

- ⭐ **Direction: inverted in aVR** (and may be inverted in **V1–V2**); **upright in the others**.
- Amplitude, duration and shape are listed but left blank on the slide. - *[added]* T < 5 mm in limb leads and < 10 mm in chest leads; asymmetric. - *[added]* **Peaked** = hyperkalaemia, hyperacute MI. **Deep symmetric inversion** = ischaemia, Wellens, PE (V1–V4), RVH strain, LVH strain, BBB (secondary).
- **Slide 120 (T-wave abnormalities): 🔎 deep T-wave inversion in V2–V6 (and I, aVL)** → anterior ischaemia / evolving anterior MI (Wellens pattern if symmetric in V2–V3).



Slide 120: T-wave inversion

> 🔁 **Recap: QRS, ST, T.** QRS ≤ 0.10 (lecturer 0.08), named QS/qR/rS…. ≥ 0.12 = BBB: RBBB M in V1; LBBB W in V1, M in V6. Pathological Q = old MI. ST normally iso-electric; ↑ in ≥ 2 contiguous leads = STEMI (inferior II/III/aVF → RCA, add V4R; septal V1–V2; anterior V3–V4; lateral I/aVL/V5–V6; posterior V7–V9) with reciprocal ↓. ↓ = ischaemia/NSTEMI. T upright except aVR (± V1–V2); peaked → K^+; deep inversion → ischaemia.

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3. High-Yield Comparison Tables

Table A: Normal Values (Slides 8, 17, 19, 68, 84, 91)

Item	Normal
Paper	25 mm/s · small square 0.04 s · big square 0.2 s · 300 big squares/min
Calibration	10 mm = 1 mV
SA / AV / ventricular rates	60–100 / 40–60 / 20–40
Axis	−30° to +90°
P	≤ 2.5 mm tall · ≤ 0.10 s wide · upright I/II, inverted aVR
PR	0.12–0.20 s (3–5 small squares)
QRS	≤ 0.08–0.10 s
QTc	≤ 0.44 s (QT/√RR)
ST	Iso-electric
T	Upright except aVR (± V1–V2)

Table B: Heart-Rate Methods

Method	Rhythm	Formula	Accuracy
R–R big squares	Regular	300 ÷ big squares	Good

Method	Rhythm	Formula	Accuracy
R–R small squares	Regular	1500 ÷ small squares	**Best**
Triplicate	Regular	300-150-100-75-60-50	Quick
6-second	**Irregular**	QRS in 30 big squares × 10	**Least accurate**

Table C: Axis

	Lead I	aVF (II)	Causes *[added]*
Normal	+	+	—
LAD	+	−	LAFB, inferior MI, LVH, WPW
RAD	−	+	RVH, PE, COPD, lateral MI, LPFB
Extreme	−	−	VT, severe hyperkalaemia, lead reversal

Table D: Atrial Enlargement

	RAE (P pulmonale)	LAE (P mitrale)
Lead II	**Tall, peaked > 2.5 mm**	**Wide > 0.10 s, notched (> 40 ms between peaks)**
V1	Tall initial positive part	**Deep, wide terminal negative part (> 0.04 s)**
Causes *[added]*	Pulmonary HTN, COPD, PE, TR	MS, MR, LVH, HTN

Table E: STEMI Territories *[added; consistent with slides 109–116]*

Leads with ST↑	Territory	Artery	Reciprocal
II, III, aVF	Inferior	**RCA** (80%) / LCx	I, aVL
+ **V4R**	Right ventricle	Proximal RCA	—
V7–V9 (ST↓ + tall R V1–V3)	Posterior	RCA / LCx	V1–V3
V1–V2	Septal	LAD (septal)	(inferior)
V3–V4	Anterior	LAD	Inferior
I, aVL, V5–V6	Lateral	LCx / diagonal	Inferior
V1–V6 + I, aVL	Extensive anterior	**Proximal LAD**	Inferior

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4. Rules, Exceptions, Patterns

Rule	Exception / nuance
Big complexes = LVH	Check **calibration**: double standard doubles everything
Small complexes = low voltage	Half standard halves everything; real causes include effusion, obesity, COPD, hypothyroid
P upright everywhere	**Inverted in aVR** is normal; inverted in I suggests lead reversal/dextrocardia *[added]*
One P per QRS	Extra P waves = **2nd/3rd-degree block**; P after QRS = **AVNRT/junctional**
PR long = block	Short PR + delta wave = **WPW**
QRS ≤ 0.10 s	≥ 0.12 s = BBB, ventricular rhythm, paced, hyperkalaemia
T upright	Normally inverted in **aVR**, may be in **V1–V2** (and III)

Rule	Exception / nuance
ST↑ = STEMI	Also **pericarditis** (diffuse, PR↓), **LBBB**, LVH, early repolarisation, ventricular aneurysm *[added]*
Inferior STEMI	**Always record V4R** (RV infarct: avoid nitrates, give fluids *[CAD link]*)
Rate by 300/big squares	**Irregular rhythm → 6-second × 10**

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5. Clinical Correlations

- **Hypertensive patient: S V1 + R V5 = 40 mm with lateral T inversion → LVH with strain**.
- **Mitral stenosis: P mitrale** → later **AF** (no P waves, irregularly irregular).
- **COPD / pulmonary HTN: P pulmonale + right axis deviation** (± RBBB).
- **Large pericardial effusion: low voltage** ± electrical alternans *[Pericardial lecture]*.
- **Hypotension and bradycardia in an inferior STEMI → RV infarct** (V4R ST↑) → fluids, avoid nitrates.
- **Young patient, palpitations, short PR + delta wave → WPW** → avoid AV-nodal blockers in AF.
- **Syncope with P waves marching through a slow wide escape → complete heart block** → pacing.
- **Renal failure with peaked T and wide QRS → hyperkalaemia** *[added]*.

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6. Rapid Revision (Last-Day)

1. **SA 60–100 · AV 40–60 · ventricle 20–40** (emergency 150–250).
2. **Limb: I, II, III bipolar; aVR, aVL, aVF unipolar. Chest V1–V6 unipolar.** V1–V2 septal, V3–V4 anterior, V5–V6 lateral.
3. Toward + electrode = upward deflection.
4. **25 mm/s: small square 0.04 s, big square 0.2 s; 300 big squares = 1 min; 10 mm = 1 mV.**
5. **11 steps:** ID · standardisation · voltage · axis · rhythm · rate · P · PR · QRS · ST · T.
6. **Full standard = 10 mm** (half 5, double 20).
7. **LVH: SV1/V2 + RV5/V6 ≥ 35 mm.**
8. **Axis:** normal −30° to +90°; LAD physiological 0 to −30, pathological −30 to −90; RAD +90 to +180; extreme −90 to ±180.
9. **Rhythm:** compare R–R intervals.
10. **Rate:** 300/big · 1500/small · 300-150-100-75-60-50 · irregular → 6 s × 10 (least accurate).
11. **P:** upright I/II, inverted aVR, ≤ 2.5 mm, ≤ 0.10 s. **Tall = RAE (pulmonale), wide/notched = LAE (mitrale).** Absent = AF/flutter/nodal/SVT.
12. **PR 0.12–0.20 s.** Long = block (1st > 0.20; Wenckebach progressive). Short = WPW/junctional.
13. **QRS ≤ 0.08–0.10 s.** Names: QS, qR, Qr, qRs, R, Rs, rS. **≥ 0.12 = BBB**: **RBBB rSR′ in V1 + wide S in V6; LBBB QS in V1 + notched R in V6.**
14. **QTc ≤ 0.44 s = QT/√RR.**
15. **ST iso-electric.** ↑ inferior II/III/aVF (reciprocal I, aVL); septal V1–V2; anterior V3–V4; lateral I/aVL/V5–V6; RV → V4R; posterior → V7–V9.
16. **T upright except aVR (± V1–V2).** Deep inversion → ischaemia.

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7. Exam-Style Questions

Q1. On a regular ECG the R–R interval is 4 big squares. What is the heart rate? A. 50 · **B. 75** · C. 100 · D. 150 · E. 300

> **Answer: B.** 300 ÷ 4 = **75/min** (slide 60).

Q2. An irregularly irregular rhythm has 9 QRS complexes in 30 big squares. What is the heart rate? A. 45 · B. 60 · **C. 90** · D. 180 · E. Cannot be determined

> **Answer: C.**
> - **Irregular → count in 6 s (30 big squares) × 10** = 90 (slides 61–62).
> - This is the least accurate method, but the right one for irregular rhythms.

Q3. Lead I is positive and aVF is negative; lead II is also negative. What is the axis? A. Normal · B. Physiological left axis · **C. Pathological left axis deviation** · D. Right axis deviation · E. Extreme axis

> **Answer: C.**
> - I up + aVF down = left axis. **Lead II negative** places it **beyond −30°**, so it is pathological (slide 36).
> - *[added]* Commonest cause: left anterior fascicular block.

Q4. Lead II shows a P wave 0.14 s wide with a notch 50 ms between the peaks, and V1 shows a deep terminal negative P component. What does this indicate? A. RAE · **B. LAE (P mitrale)** · C. Atrial flutter · D. Junctional rhythm · E. Hyperkalaemia

> **Answer: B** (slides 70, 73, 75).
> - *[added]* Typical of mitral stenosis.

Q5. ST elevation in II, III and aVF with ST depression in I and aVL in a hypotensive, bradycardic patient. Which extra leads should be recorded, and why? A. V7–V9 for posterior MI · **B. V4R–V6R for right ventricular infarction** · C. aVR for left main · D. None · E. Repeat in 24 h

> **Answer: B.**
> - Inferior STEMI with reciprocal change (slide 109). Hypotension and bradycardia raise suspicion of **RV involvement** → **right-sided leads** (slide 113).
> - *[added]* V7–V9 are also worth recording if V1–V3 show ST depression.

Q6. V1 shows rSR′ and V6 a wide slurred S wave; QRS is 0.15 s. Diagnosis? A. LBBB · **B. RBBB** · C. WPW · D. LVH · E. Normal variant

> **Answer: B** (slides 97, 100).

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8. Exam Pitfalls & Traps

- **Most strips in this deck have no caption.** Every reading marked 🔎 is mine. Check the starred/quiz ones (slides 54–55 have the lecturer's own answers) against your class notes.
- **Normal-axis range:** slide 36 says **0 to +90°** (with 0 to −30° as "physiological left"); slide 35 says **−30° to +90°**. Both mean the same: up to −30° is **not** pathological.
- **QRS duration:** slide 91 says **0.08 s (2 small squares)**; slide 19 says **0.06–0.10 s**. Only **≥ 0.12 s** defines a bundle branch block.
- **Step order:** the slides list "Rhythm then Rate" on some slides and "Rate then Rhythm" on others. Be consistent in the exam.
- **Calibration first:** half or double standard fakes low voltage or LVH.
- **The 6-second method is the least accurate.** Use it only for irregular rhythms.
- **P waves:** inverted in aVR is **normal**. Inverted in lead I suggests limb-lead reversal or dextrocardia *[added]*.
- **ST elevation is not always MI:** pericarditis (diffuse, PR↓), LBBB, LVH, early repolarisation *[added]*.
- **Inferior MI:** always look for **RV (V4R)** and **posterior (V7–V9)** extension.
- **Low-voltage criteria** were only a heading on slide 27. The cut-offs given here are standard textbook values *[added]*.